Biology-1: Lecture 03

(BIO103)

Chapter 1. Introduction to Biology

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Biological Classification

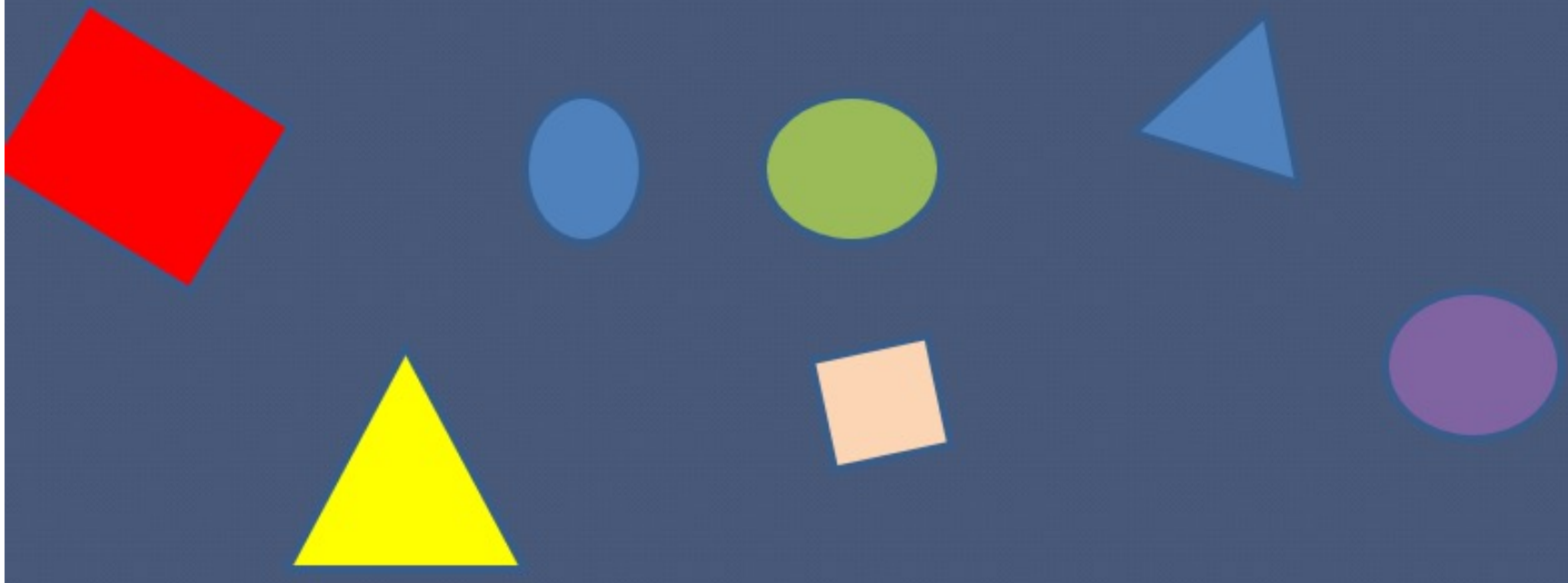
Lecture 3: Outlines **Spring 2026**

1. **Biological Classification**

   - Two-kingdom classification
   - Three-kingdom classification
   - Four-Kingdom Classification
   - Five-kingdom classification
   - Phylogenic system classification

Biology Classification

- It's a big confusing mess out there
- Someone has to sort it all out…can you?



Basis of Biology Classification

- Systematics (current classification system) uses the following to classify
  - cell type
  - how food and energy are obtained
  - structure and function of features
  - common ancestry
  - molecular analysis (DNA)

Biological classification

- The grouping of similar things for a specific purpose is called **classification.**

- In biology, classification is the process of **arranging organisms into groups based on similar characteristics**.

- The science of naming and classifying organisms is called **taxonomy**.

- The term is derived from the **Greek taxis ("arrangement**") and **nomos ("law**").

- Taxonomists classify organisms into a structural hierarchy—a multi-level system in which each group is nested, or contained, within a larger group.

Early Biological classification

- The earliest humans probably classified **organisms as plants and animals.**

- They may have further classified **plants as edible or poisonous**, and the **animals as harmful or harmless**.

- The first serious attempt was made **at 300 BC** to classify all the organisms by the Greek philosopher and scientist, **Aristotle and his students**.

Aristotle's Biological classification

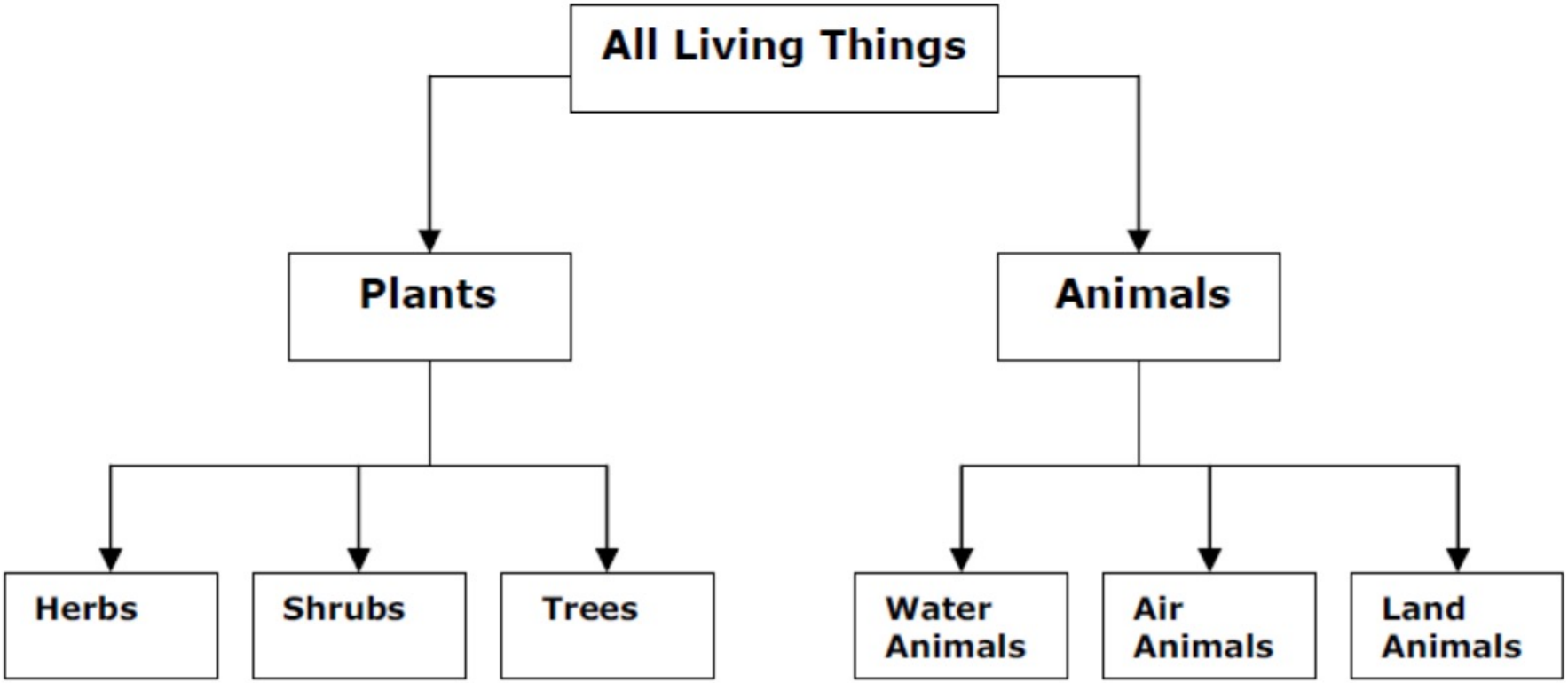
- Aristotle and his students first classified the **organisms as plant or animal.**

- They then **classified the animals** according to **where they lived,** resulted in three groupings**: air animals, water animals, and land animals**.

- They classified the **plants according to the structure of stems.**

- Those with soft stems were called herbs; those with a **single woody stem were called trees**; and those with many small woody stems were called shrubs.

Aristotle's Biological classification



Aristotle's Biological classification

- Aristotle's classification system survived for almost two thousand years.
- However, by the beginning of the 18th century, over 10,000 kinds of organisms were known and Aristotle's system was unable to classify them all.
- Many of newly discovered organisms would not fit into any category of Aristotle's simple system.

Modern Biological Classification

- **Carolus Linnaeus**, a Swedish botanist, developed a simple classification system that **forms the basis of our modern method of classificatio**n.

- At the start of the 18th century about 10,000 kinds of organisms were known. By the end of that century over 70,000 kinds were known.

- Linnaeus tried to develop a classification system for this large number of organisms.

- **By 1753** his system was well developed and **modern taxonomy began.**

Carolus Linnaeus: Binomial Nomenclature

- Once Linnaeus had decided on a basis for classifying organisms, he then **developed a system for naming them**.
- He gave each species a name that consists of **two words.** This system is called binomial nomenclature.
- He used Latin words for these names because **all scientists wrote in Latin in time of Linnaeus**.
- Thus, the human is ***Homo sapiens***, and the domestic (house) cat is ***Felis domesticus***.

Carolus Linnaeus: Binomial Nomenclature

- The first word of each name is called the genus and the second word is called the species.
- The genus begins with a capital letter and the species does not.
- The genus and species are either printed in italics or underlined.

Main Classification Groups (Taxa)

- There are seven main taxa or classification groups:

1. **Species:** Species is a group of individuals that are alike in many ways and interbreed under natural conditions to produce fertile offspring (children).

2. **Genus:** Genus (plural genera) is a group of species that are closely similar in structure and evolutionary origin.

3. **Family**: Family is a group of similar kinds of genera. That is, similar genera are grouped to form a taxon called Family.

4. **Order:** Similar families are grouped to form a taxon called order.

Main Classification Groups (Taxa)

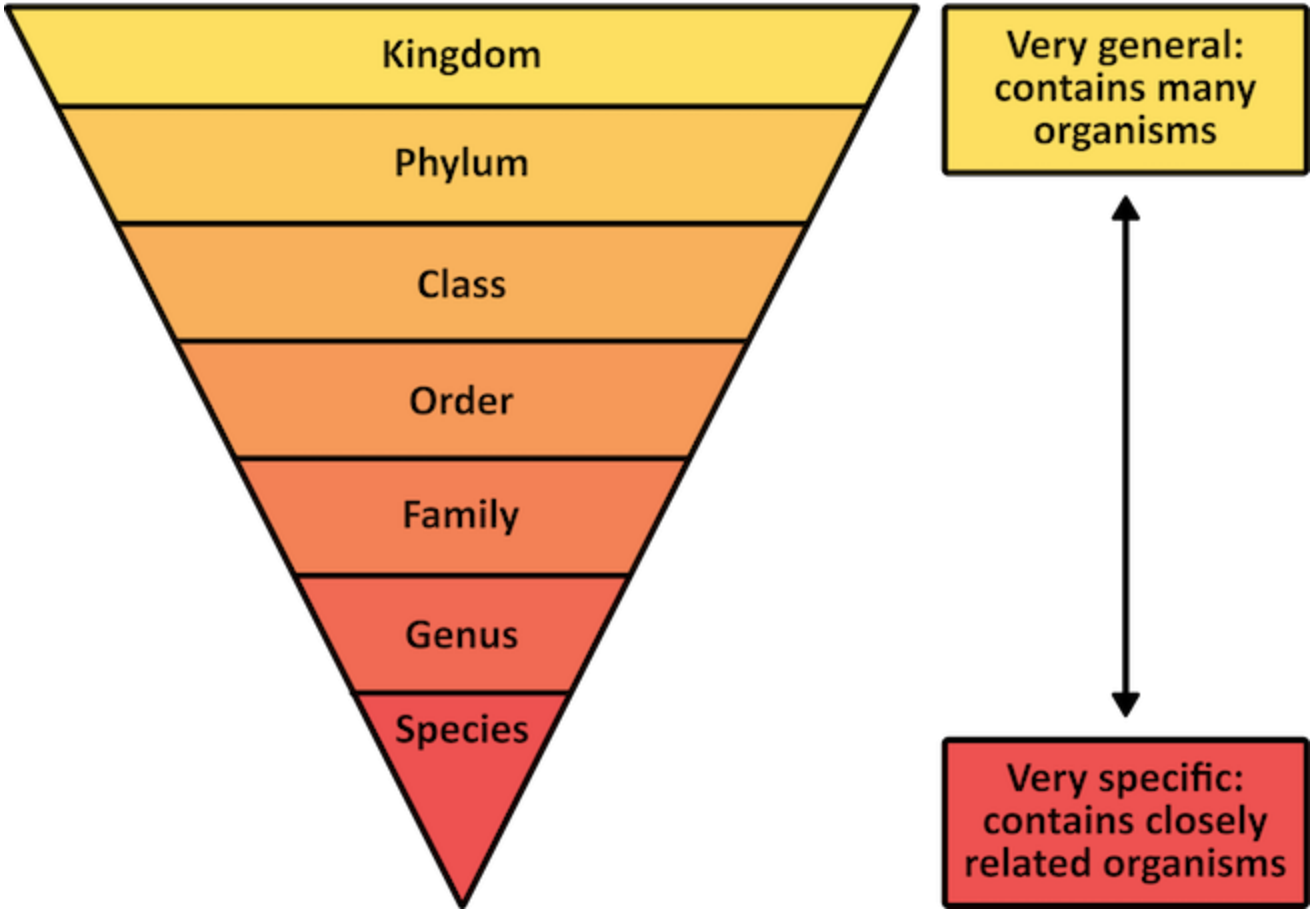
5. **Class:** Similar orders are grouped to form a taxon called class.

6. **Phylum or Division**: Similar classes are grouped to form a taxon called phylum or division. **Zoologists favour phylum** and **botanists favour division.**

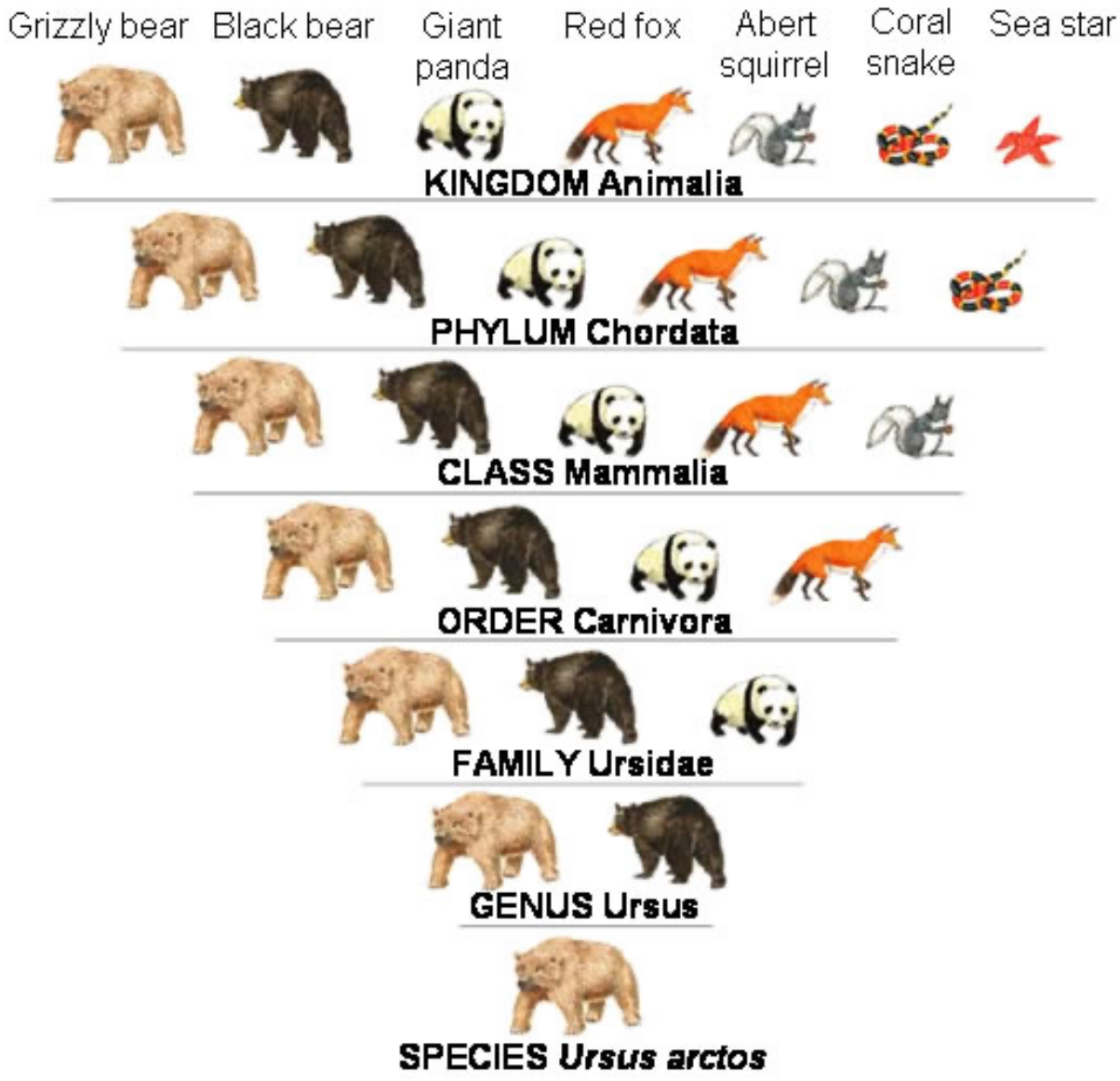
7. **Kingdom**: All the phyla or divisions that contain animals are grouped in the kingdom **Animalia**, and all the phyla or divisions that contain plants are grouped in the kingdom **Plantae**.

Modern Classification System

King Philip Can Only Find Green Socks



Modern Classification System



Main Classification Groups (Taxa)

Classification of Some Organisms

Taxon	***Human***	***Gorilla***	***Dog***	***Cat***
Kingdom	Animalia	Animalia	Animalia	Animalia
Phylum or Division	Chordata	Chordata	Chordata	Chordata
Class	Mammalia	Mammalia	Mammalia	Mammalia
Order	Primates	Primates	Carnivora	Carnivora
Family	Homonidae	Pongigae	Canidae	Felidae
Genus	*Homo*	*Gorilla*	*Canis*	*Felis*
Species	*sapiens*	*gorilla*	*familiaris*	*domesticus*

Two-Kingdom Classification

- **Carolus Linnaeus introduced** this system of classification in 1753.

- This is the oldest and perhaps still the most widely used system of classification.

- It groups all organisms into two kingdoms, (1) Plantae, which includes bacteria, algae, fungi and higher plants, and (2) Animalia, which includes protozoa and higher animals.

- That is, all living things are assumed to be either plants or animals.

Three-Kingdom Classification

- Ernst H Haeckel introduced this system of classification in 1866.
- This system still has the kingdoms (**1) Plantae and (2) Animalia**, however, it has a third kingdom, **(3) Protista**.
- The kingdom **Protista contains** all microorganisms, including the bacteria, blue-green algae, protozoa, unicellular algae (eg., flagellales), and microscopic fungi.
- This system was recognized by many biologists during the first half of the 20th century and is still very popular.

Four-Kingdom Classification

- To solve the problem with prokayotic and eukaryotic organisms in classification, Herbert Faulkner Copeland (1902- 1968) proposed the four kingdom classification in 1956.

- The four kingdoms were **Monera, Protista, Plantae, and Animalia**.

- He grouped **unicellular organisms** into two large kingdoms: the Monera kingdom and the Protista kingdom.

Four-Kingdom Classification

- In 1966, he included bacteria and one of the most primitive algae, called blue- green algae, under the Monera kingdom.
- The kingdom Monera consists of unicellular organisms.
- The kingdoms Protista, Plantae, and Animalia are the eukaryotic organisms.
- The **fungi were placed in the kingdom Plantae** which was one of the drawbacks of the four kingdom classification.

Five-Kingdom Classification

- **Robert H Whittaker** introduced this system of classification in 1969.

- In this system **fungi were separated from the kingdom Plantae** as they are quite different from the usual plants.

- Therefore, the five kingdoms are (**1) Plantae, (2) Animalia, (3) Fungi (4) Protista, and (5) Monera**.

Phylogenetic System Classification

- **Carl R Woese** introduced a completely new approach in classification of organisms in 1977 that is based on evolutionary (phylogenetic) relationship among organisms.

- Woese meticulously analysed **ribosomal ribonucleic acid (rRNA)**, genetic molecules that coordinate part of protein production.

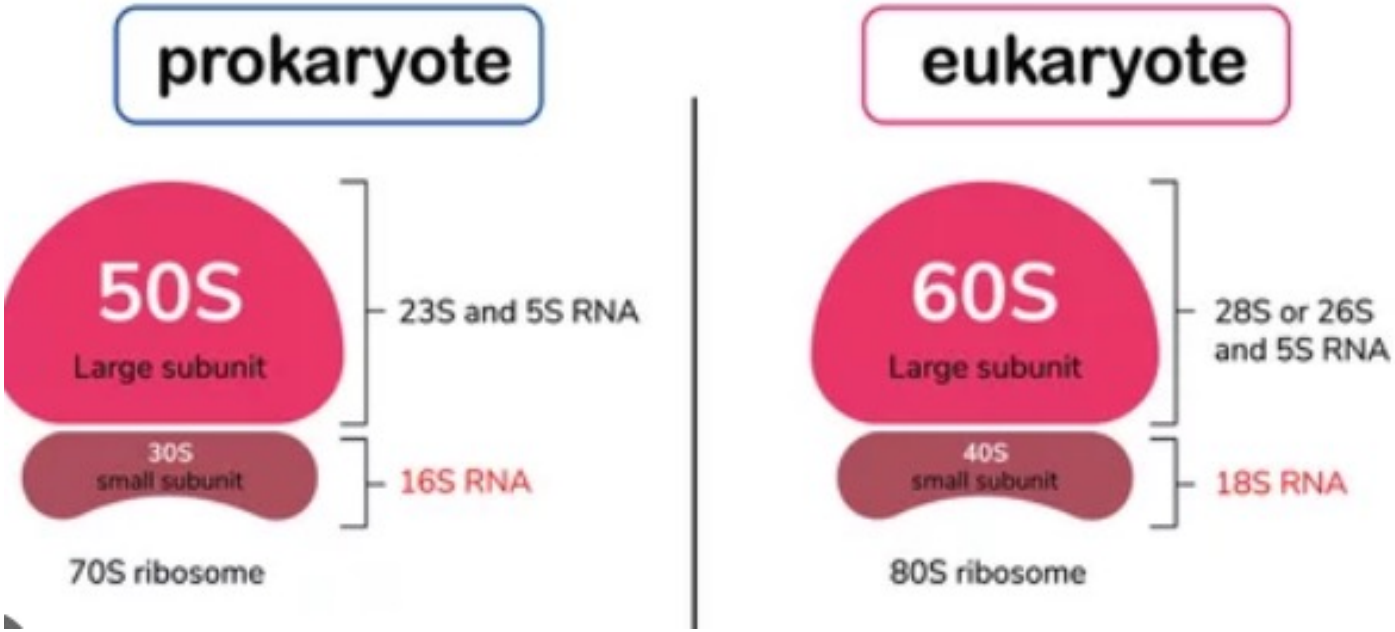
- Because rRNA shows only slight variation from one generation to the next, it is an excellent tool for revealing the evolutionary, or family, relationships among organisms.

Phylogenetic System Classification

- There are three important RNAs occurring in the cells – rRNA, mRNA, and tRNA (transfer RNA).
- rRNA or ribosomal RNA are molecules found in the cells involved in the protein synthesis of ribosomes with the help of mRNA (messenger RNA).
- 16S rRNA is a type of rRNA involved in making the small subunit of the prokaryotic ribosome; these are components of the 30S subunit (prokaryotic ribosome).

In prokaryotes, the ribosomal RNAs are as follows –

Name	Where can it be seen?	Size
5S	Large subunit of ribosome	120 nucleotides
16S	Small subunit of ribosome	1500 nucleotides
23S	Large subunit of ribosome	2900 nucleotides

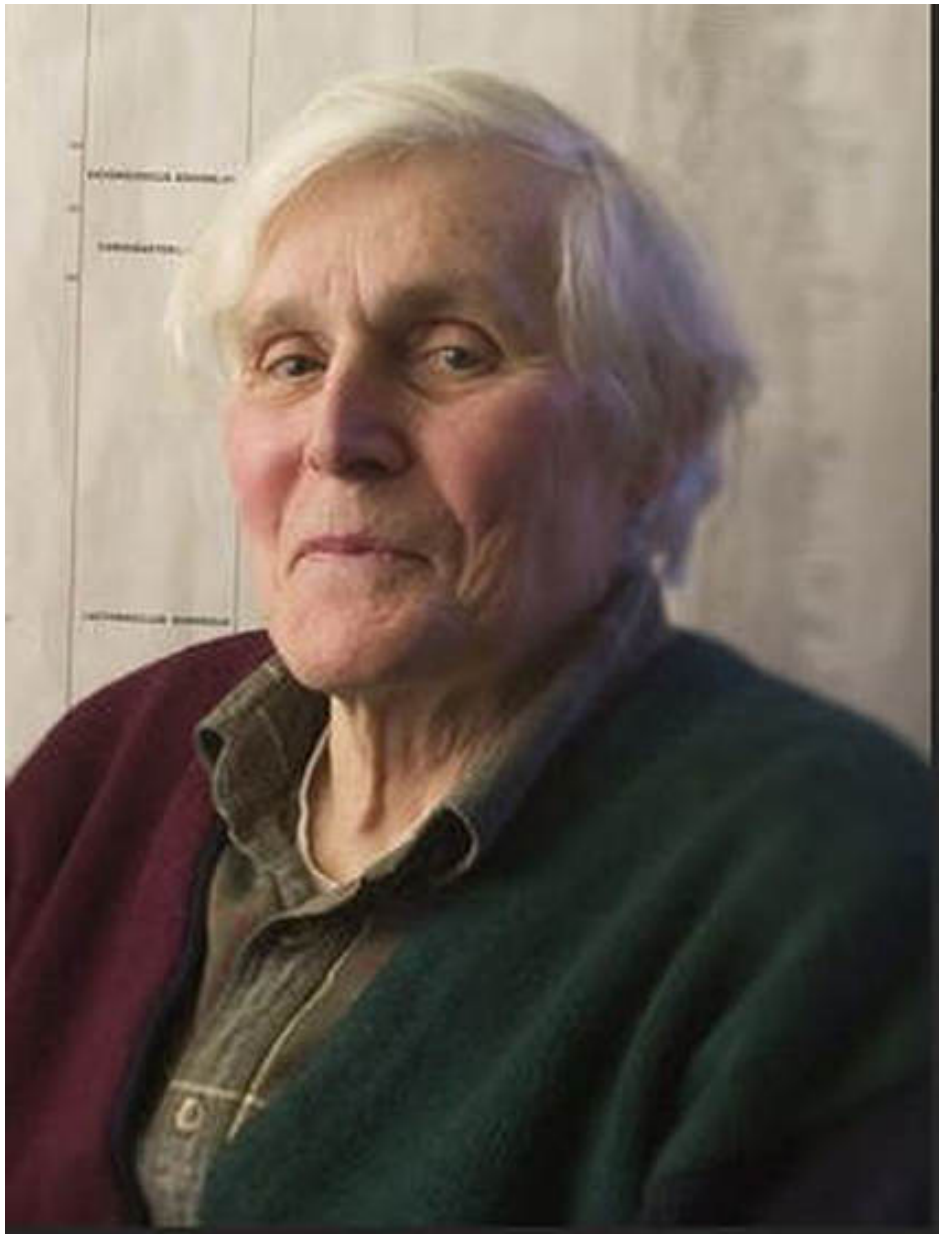


Phylogenetic System Classification

- Woese spent nearly a decade analysing the rRNA of various types of bacteria and arranging them into a microbial evolutionary tree.

- Use this concept all organisms are now classified into three kingdoms.

- They are **(1) Archaea, (2) Bacteria, and (3) Eukarya.**



Carl R. Woese (1928-2012), professor of microbiology and a faculty member of the Institute for Genomic Biology at the University of Illinois at Urbana-Champaign.

The portrait was taken by Don Hamerman.

The photo was taken in 2004 at the University of Illinois.

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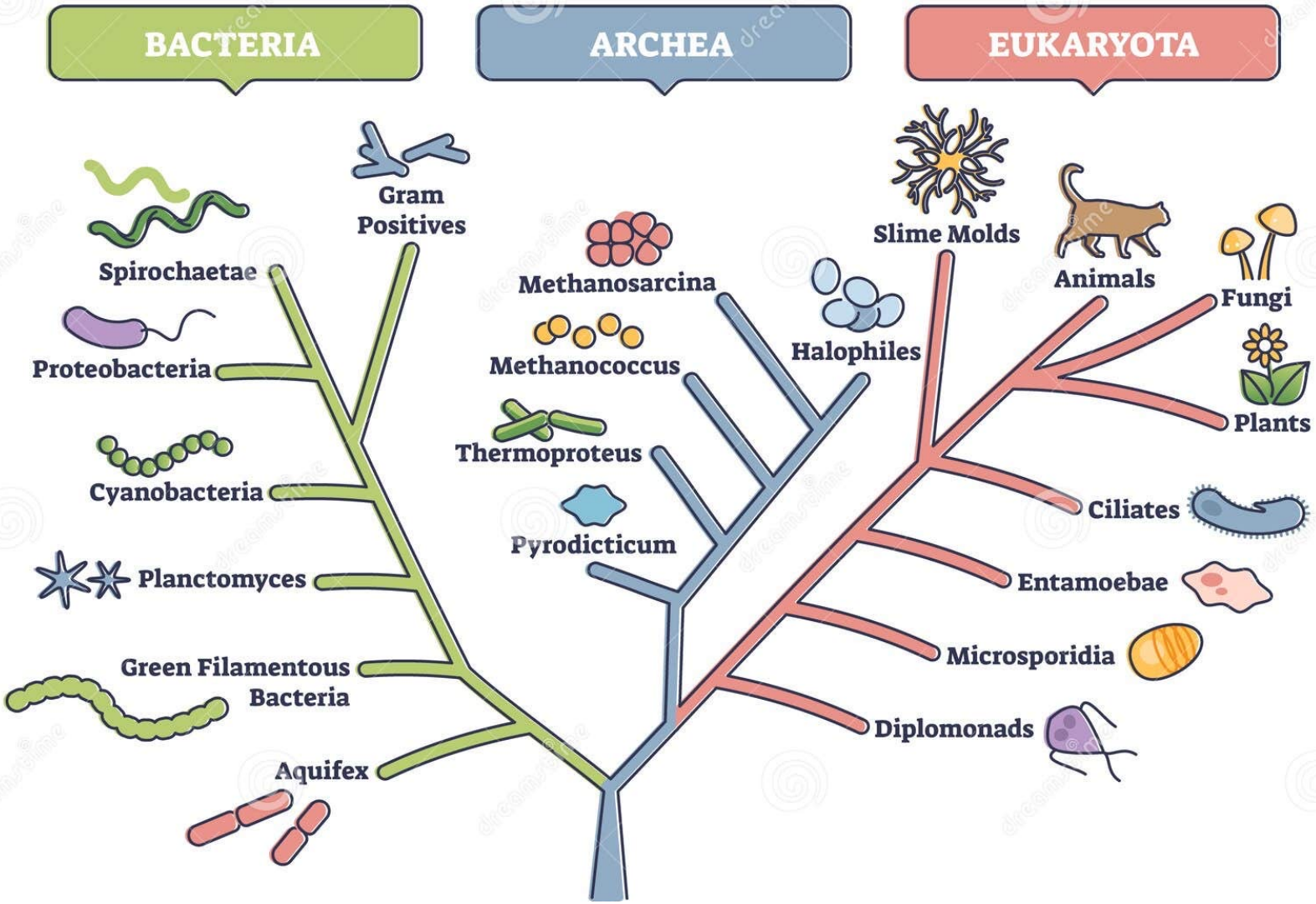
PHYLOGENETIC TREE



Summary of Various Classification Systems

Classification	*System Kingdom*	*Organisms Included*
2-Kingdom System *Carolus Linnaeus, 1753*	(1) Plantae	Bacteria, Cyanobacteria, Unicellular algae, Multicellular algae, Fungi & Higher plants
	(2) Animalia	Protozoa & Higher animals
3-Kingdom System *Ernst H Haeckel, 1866*	(1) Plantae	Multicellular algae, Fungi & Higher plants
	(2) Animalia	Higher animals
	(3) Protista	All microorganisms: Bacteria, Cyanobacteria, Unicellular algae, & Protozoa
4-Kingdom System	(1) Plantae	Multicellular algae, Fungi & Higher plants
	(2) Animalia	Higher animals
	(3) Protista	Eukaryotic microorganisms: Unicellular algae & Protozoa
	(4) Monera	Prokaryotic microorganisms: Bacteria, & Cyanobacteria
5-Kingdom System *Robert H Whittaker, 1959*	(1) Plantae	Multicellular algae & Higher plants
	(2) Animalia	Higher animals
	(3) Fungi	All fungi: Yeasts & Molds
	(4) Protista	Eukaryotic microorganisms: Unicellular algae & Protozoa
	(5) Monera	Prokaryotic microorganisms: Bacteria, & Cyanobacteria
Phylogenetic System *Carl R Woese, 1977*	(1) Archaea	Archaebacteria
	(2) Bacteria	Bacteria
	(3) Eukarya	Eukaryotic organisms: Algae, Fungi, Higher plants, Protozoa & Higher animals